

## RESEARCH ARTICLE

# Influence of PP types on migration of zinc from nano-ZnO/PP composite films

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Polypropylene nanocomposites have great potential applications in food packaging and the evaluation of the migration of nanoparticles is important for the food safety. Three types of polypropylene (homopolymer of polypropylene, PPH; blockcopolymer of polypropylene, PPB; random copolymer of polypropylene, PPR) nanocomposite films with or without coupling agent were tested in 3% acetic acid as food simulant. The migration amount of zinc was quantified by inductively coupled plasma atomic emission spectroscopy. The surface morphology and crystallinity were explored by field emission scanning electron microscopy, atomic force microscopy, and differential scanning calorimetry, respectively. The results showed that the incorporation of titanate coupling agent could inhibit ZnO-NPs migration from the films since titanate coupling agent could improve the interface between ZnO-NPs and PP molecular chain segment. Furthermore, the migration rate of zinc was dependent on the degree of crystallinity ( $X_c$ ) of PP and their order of the maximum migration rate was PPH, PPB and PPR. The notable phase separation could be observed from the morphology of nano-ZnO/PPH, while ZnO-NPs could be well dispersed in the PPR matrix. Thus, ZnO-NPs restricted the molecular chain segment of PPH and reduced its  $X_c$  from 88.6% to 73.2%. The results demonstrated that titanate coupling agent and PPR could inhibit the migration of ZnO-NPs significantly.

**KEYWORDS**

coupling agent, material structure, migration, nanoparticle, polypropylene, zinc oxide

## 1 | INTRODUCTION

Recent advances have demonstrated that food packages produced with nanomaterials have been more available in the current market.<sup>1,2</sup> Among them, the zinc oxide nanoparticles (ZnO-NPs) reveal antimicrobial properties and improve packaging properties, such as mechanical strength, barrier properties, and ultraviolet properties.<sup>3-5</sup> However, the researches indicate that ZnO-NPs cause toxicity to bacterial. Their potential mechanisms of toxicity are complex and have not been clearly interviewed.<sup>6</sup> The migration of nanoparticles into food is usually regarded as the main factor related to food safety<sup>7</sup> whereas only a few studies are about the migration of ZnO-NPs. Panea et al<sup>8</sup> reported that the migration of nano-ZnO (10% w/w) from LDPE films into the aqueous food simulant was below the detection limit ( $<0.005 \text{ mg}\cdot\text{kg}^{-1}$ ) analyzed by inductively coupled plasma mass

spectrometry (ICP-MS). In our group, the detection methods of ZnO in nano-ZnO/LDPE film and food simulant were established by ICP-atomic emission spectroscopy (AES) and ICP-MS, respectively.<sup>9</sup> Nevertheless, any specific legislation for nano-ZnO has not been clearly defined until now. It is also worth to note that the regulation (EU) No. 2016/1416 has updated the specification of specific migration limit for zinc, which reduced from 25 to  $5 \text{ mg}\cdot\text{kg}^{-1}$  food or food simulant.<sup>10</sup>

Polypropylene and its copolymers (unified called PP here) are extensively used in food packaging.<sup>11</sup> PP can be classified into three different types as homopolymer of polypropylene (PPH), blockcopolymer of polypropylene (PPB), and random copolymer of polypropylene (PPR). Based on previous studies<sup>12-14</sup> and our groups' researches,<sup>15,16</sup> not only temperature and immersion time but also different types of PP matrix make significant influence on the

migration of additives. It has been observed that the migrations of the common additives, such as antioxidants, stabilizers, and ultraviolet absorbent agents, from PPR are higher than those from PPB, and the lowest migration was observed from PPH. These existing researches are aimed at various organic chemical compounds with low molecular weight. The discrepant migration behaviors of the additive from different types of PP are related to the crystallinity of the material. The higher the crystallinity, the more unfavorable the migration of the additive. This paper plans to discuss whether the effects of different types of PP on the nano-components are consistent with the migration rule of organic additives. That is of great significance to further explore the influence of PP material type on migration.

The ZnO-NPs have a strong tendency to agglomerate due to its high surface energy, so it should be modified by coupling agent as titanium triisostearoylisopropoxide to improve miscibility and dispersibility. In this study, ZnO-NPs in unmodified and modified states were separately incorporated into three types of PP matrix by using melt mixing and blowing process to manufacture the nano-ZnO/PP composite films. Acetic acid (3% w/v) was chosen as food simulant and the migration tests in different condition were performed to discuss the migration behavior of ZnO-NPs. Moreover, the characterizations of nano-ZnO/PP composite films were observed by using scanning electron microscope (SEM), atomic force microscope (AFM), and differential scanning calorimetry (DSC), respectively. The study could promote the evaluation for the migration of nanoparticles of metal oxide from PP films. In the view of safety, PPH could not be used as matrix of nanoparticle/PP composite for food packaging.

## 2 | EXPERIMENTAL

### 2.1 | Materials and methods

#### 2.1.1 | Reagents and chemicals

PPH (Z30S; MFR 2.5-3.5 g/10 min (ASTM D1238, VST 230°C); Tacticity  $\geq 96\%$ ), PPB (EPS30R; MFR 1.2-1.8 g/10 min (ASTM D1238, VST 230°C)) and PPR (5090 T; MVR 1.5 cm<sup>3</sup>/10 min (ISO 1133, VST 230°C)) were supplied in pellet (Chinese Sinopec Zhongyuan Petroleum Chemical Co., Ltd, Henan, China). ZnO-NPs (50  $\pm$  10 nm) and titanium triisostearoylisopropoxide (HY13B) were purchased from Aladdin Industrial Corporation (Shanghai, China). An electronic analytical balance (FA1604N; Jinghai Instrument Co., Ltd, Shanghai, China) was used. 65% nitric acid (HNO<sub>3</sub>) (Guaranteed Reagent), 30% hydrogen peroxide (H<sub>2</sub>O<sub>2</sub>) (Guaranteed Reagent), and glacial acetic acid (Analytical Reagent) were obtained from Guangzhou Chemical Reagent Co. (Guangzhou, China). The standard solution of zinc (1000  $\mu\text{g}\cdot\text{mL}^{-1}$ ) was purchased from Aladdin Industrial Corporation (Shanghai, China). Stock solution of 20  $\mu\text{g}\cdot\text{mL}^{-1}$  was stored at 4°C. The solutions were prepared with ultrapure water from a Milli-Q water purification system (Advantage A10; Millipore, Boston, MA, USA).

#### 2.1.2 | Samples preparation

The surface modification of ZnO-NPs was performed as follows: the titanate coupling agent (1.50 g) and isopropyl alcohol (320.0 mL) were

mixed and stirred, followed by stirring with a constant temperature (30°C) and a stable speed (500 r·min<sup>-1</sup>) by the electromagnetic stirrer (Shanghai Analysis Instrument Co., Ltd, Shanghai, China) for 1 hour. Then, the ZnO-NPs (50.00 g) were added into the mixture, and the whole mixture was further stirred for 1.5 hours. At last, the resulting slurry was dried by the vacuum drying oven at 85°C for 6 hours. The dried powders were then crushed and sealed in a glass desiccator.

#### 2.1.3 | Nanoparticle composite films

The films were manufactured by the twin screw extrusion granulation equipment (MEDI-22/40; POTOP Experimental Analysis Instrument Co., Ltd, Guangzhou, China) and the film-blowing machine (JFYC; Jinfang yuan Machinery Manufacturing Co., Ltd, Guangzhou, China). Unmodified and modified ZnO-NPs were separately added into blank PPH, PPB, and PPR pellets. After being well mixed, they were used to produce new pellets by the twin extrusion granulation equipment with processing temperatures ranging from 190°C to 200°C and screws speed of 20.0 r·min<sup>-1</sup>. Then, the extrudates were cut into 2.0-mm-length pellets by a pelletizer (AGVU-04; POTOP Experimental Analysis Instrument Co., Ltd, Guangzhou, China). The aforementioned process was repeated for three times. The composite pellets were then converted into films by the film-blowing machine, which was carried out at a temperature range from 200°C to 205°C, a screw speed of 35.0 r·min<sup>-1</sup>, and a take-off speed of 14.0 r·min<sup>-1</sup>. The average thickness of the blow-extruded films was 32  $\pm$  2.0  $\mu\text{m}$ . There were nine types of polymer films manufactured altogether in this study: three types of blank PP films were termed b-PPH, b-PPB, and b-PPR, respectively; three types of nanocomposite PP composite films containing 2% (w/w) unmodified ZnO-NPs were termed ZnO-PPH, ZnO-PPB, and ZnO-PPR, respectively; and three types of nanocomposite PP composite films containing 2% (w/w) modified ZnO-NPs with couple agents were termed ZnO-PPH#, ZnO-PPB#, and ZnO-PPR#, respectively.

### 2.2 | Initial concentration of zinc in samples

Before the migration experiment, the actual initial loading of ZnO-NPs in the films were quantified. All types of films (b-PPH, b-PPB, b-PPR, ZnO-PPH, ZnO-PPB, ZnO-PPR, ZnO-PPH#, ZnO-PPB#, and ZnO-PPR#) were cut into small pieces. The nanocomposite films were digested using a microwave digestion system (ETHOS CMMWG; Milestone, Bergamo, Italy) equipped with 10 polytetrafluoroethylene vessels. Portions of the films (0.100 g) were weighed into a polytetrafluoroethylene vessel. After that, a mixture of 5-mL HNO<sub>3</sub> and 2-mL H<sub>2</sub>O<sub>2</sub> was added; the vessels were put into the microwave digestion chamber. The samples could be better digested by using the optimized procedure as follows: the temperature was increased to 160°C in 10 minutes and was subsequently increased to 220°C in 5 minutes with power of 1000 W and pressure of 3.5 MPa, and held at 220°C for another 30 minutes. After a cooling time of 2 hours, each solution in the vessel was soluted with ultra pure water to 100.0 mL before analysing it by ICP-AES (ICAP 6500; Thermo Electron Corporation, Boston, USA). The conditions of ICP-AES have been described in previous studies.<sup>9</sup> Each sample was analysed in triplicate.

## 2.3 | Migration tests

The migration tests were carried out according to the European Union regulation No 10/2011<sup>17</sup> for plastic materials and articles intended to be in contact with food. According to pre-experiment and previous studies,<sup>18,19</sup> the migration amount of nano-metal or nano-metal oxide (eg, Ag, Cu, TiO<sub>2</sub>) from polyolefin into aqueous acetic acid was much higher than that into aqueous ethanol and water, so 3% acetic acid was chosen as food simulant in order to show more clearly the effect of different types of PP on release of nanoparticles; 15 mL of 3% acetic acid was placed in a centrifuge tube (50 mL) with one piece (3.0 cm × 3.0 cm, approx. 0.027 g) of a nanocomposite film sample. The samples were left for specific time intervals at specific temperatures in a water bath kettle (Gongyi Instrument Co., Ltd, Henan, China). In order to investigate the migration behavior of ZnO-NPs, the effects of three migration temperatures (20°C, 40°C, 70°C), immersion time, three types of PP (PPH, PPB, PPR), and two kinds of composite (with or without coupling agent) were investigated. The determination of zinc in 3% acetic acid was carried out by ICP-AES. Each analyzed sample was measured in triplicate with a standard deviation below 5%.

Migration results were normalized as migration rate which was expressed in terms of the ratio between the amount of zinc migrating into the food simulants and the actual initial amount of zinc in nano-ZnO/PP composite films.

## 2.4 | Migration under cycle use conditions

In order to study the influence of three types of PP (PPH, PPB, and PPR) on the release of ZnO-NPs, cycle migration test was performed. Each sample was soaked in the simulant of 15 mL of 3% acetic acid in a centrifuge tube at 70°C for 2 hours. Then, the simulant was directly analyzed by ICP-AES without any treatment. This was the first cycle of the experiment. That sample film was removed immediately to another centrifuge tube for the repeated second cycle of the experiment. Afterwards the third cycle was performed. The exposure experiments were named in sequence as cycle 1, cycle 2, and cycle 3. Total migration (unit mg kg<sup>-1</sup>) is the sum of the amount of migration in the film detected in the three cycles. It is intended to be compared with the initial concentration.

## 2.5 | Characterization of nanocomposite films

The original morphology of ZnO-NPs and the appearance of nano-ZnO/PP composite films were observed by ULTER55FE-SEM (Germany Zeiss Jena Carl Inc., Germany). All films were cleaned with anhydrous ethanol and dried in air before cutting into 25-mm<sup>2</sup> square segments. For SEM, the dry segments were connected to copper cylinder by conductive paper and gold sputtered to produce electric conductivity. AFM (BRUKER Optics Instrument Co., Ltd, Saarbrücken, Germany), equipped with Tap 150Al-Gprobe, was used to survey the surface morphology of different types of nanocomposite films. The surface roughness of the film samples was measured by AFM with a constant force of 5 N m<sup>-1</sup> and a resonance frequency of 150 kHz. The roughness average ( $R_a$ ) and the root mean squared roughness ( $R_q$ ) were determined from collected images 25 μm × 25 μm in size.

The experiment should be repeated four times. Surface roughness was analyzed by Nano Scope Analysis 1.40 (Bruker Corporation, Germany).

## 2.6 | DSC analysis

Differential scanning calorimeter (DSC 204F1 Phoenix, NETZSCH-Gerätebau GmbH, Germany) measurements were carried out as following: the blank PP and nano-ZnO/PP were tested under a nitrogen flow rate of approximately 50 mL·min<sup>-1</sup>. They were heated from 25°C to 180°C at the rate of 10°C min<sup>-1</sup> and kept at the high temperature for 2 minutes in order to erase the previous thermo mechanical history and to obtain a completely relaxed melt. Then, the melt was cooled to 25°C in the air and held at room temperature for 1 minute. Each sample was subjected to two heating/cooling cycles, and data were recorded during the heating.

## 2.7 | Statistical analysis

Minitab 17.0 analytical software (Minitab Co., State College, USA) was used to carry out the statistical analysis; 95% confidence level is used, and  $P < 0.05$  means there is a significant difference between two sets of data.

# 3 | RESULTS AND DISCUSSION

## 3.1 | Initial concentration of the zinc in samples

From Table 1, the actual zinc content of the nanocomposite films was less than the added amount of ZnO-NPs due to its loss during the manufacture processing. The loss of ZnO-NPs was somewhat large is attributed to the strong electrostatic attraction and the weak compatibility between ZnO and PP. In addition, the amount of zinc in the nanocomposite films with the addition of the coupling agent was significantly higher than that without the coupling agent ( $P < 0.05$ ). The coupling agent could improve the compatibility of the nano-ZnO/PP composite films. Meanwhile, the loss of ZnO-NPs from PPH was higher than PPB and PPR, indicating that different types of PP may influence the incorporation/migration of ZnO-NPs into/from the

**TABLE 1** Initial concentration of zinc for different types of PP films ( $n = 6$ )

| Samples  | Added Concentration (g·kg <sup>-1</sup> ) | Initial Concentration (g·kg <sup>-1</sup> , Mean ± S.D.) |
|----------|-------------------------------------------|----------------------------------------------------------|
| b-PPH    | 0.00                                      | 0.00                                                     |
| ZnO-PPH  | 20.02                                     | 10.54 ± 0.17 <sup>a</sup>                                |
| ZnO-PPH# | 20.04                                     | 11.61 ± 0.16 <sup>b</sup>                                |
| b-PPB    | 0.00                                      | 0.00                                                     |
| ZnO-PPB  | 20.05                                     | 12.46 ± 0.16 <sup>a</sup>                                |
| ZnO-PPB# | 20.03                                     | 13.07 ± 0.06 <sup>b</sup>                                |
| b-PPR    | 0.00                                      | 0.00                                                     |
| ZnO-PPR  | 20.03                                     | 11.09 ± 0.74 <sup>a</sup>                                |
| ZnO-PPR# | 20.01                                     | 12.15 ± 0.11 <sup>b</sup>                                |

<sup>a,b</sup>Note:  $t$ -test between with and without coupling agent in the same type of PP films,  $P < 0.05$ .

composite films. In this study, the actual content of zinc is set to the initial concentration for the following calculations.

### 3.2 | Migration of zinc into food simulants

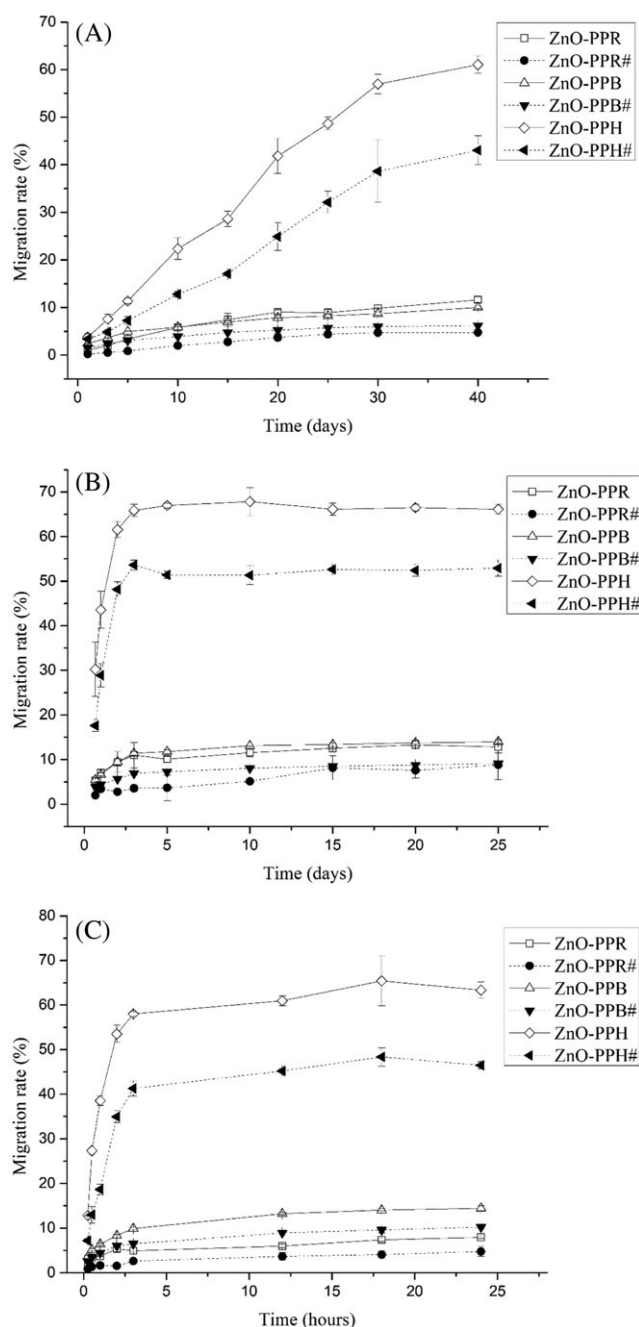
The migration rates of zinc from different types of nano-ZnO/PP films into acetic acid were shown in Figure 1. With immersion time passed, the amount of migration increased and eventually reached equilibrium. Higher temperature was favorable for the migration. For all samples, the equilibrium time at the same temperature was substantially consistent, that is, 12 hours and 5 days to reach equilibrium at 70°C and 40°C, respectively, while at 20°C, the migration still didn't reach equilibrium after a long period exposure of 40 days. With the temperature

not only decreases the equilibrium time but also increases the migration amount of zinc.

Figure 1 also showed that the migration rates from PPH samples in all cases were much higher than those from PPB and PPR ( $P < 0.05$ ). The migration rate from ZnO-PPH, ZnO-PPB, and ZnO-PPR film when reaching equilibrium was 61.0%, 10.0%, and 9.0% at 20°C, 66.2%, 13.7%, and 12.9% at 40°C, and 63.2%, 13.9%, and 7.1% at 70°C, respectively. The migration rate was in the order as PPH > PPB > PPR at 40°C and 70°C ( $P < 0.05$ ), while no significant difference between PPB and PPR was observed at 20°C. This phenomenon was unexpected and totally opposite to the previous studies on organic additives.<sup>13–15</sup> For instance, the migration of stabilizers, such as BHT, UV-P, and UV-327, decreased in the order as PPR, PPB, and PPH.<sup>16</sup> To our best knowledge, this may be the first time to discuss the migration of nano-metallic oxide from PP composite films and the variation by the PP type. The wonderful finding indicated that the migration mechanism of ZnO-NPs may be different from those organic chemical compounds with low molecular weight. According to the latest regulation of (EU) No. 2016/1416,<sup>10</sup> the maximum amount of migration from PPH had exceeded the specific migration limit of zinc.

An additional point should be mentioned in case Figure 1 that is the effect of the coupling agent on the migration of zinc from the nano-ZnO/PP composite films. There was a significant difference of the migration between samples with and without the titanate coupling agent from the same PP type. For example, at 70°C, the maximum migration rate of zinc from ZnO-PPH, ZnO-PPB, and ZnO-PPR into 3% acetic acid was 63.2%, 13.9%, and 7.1%, respectively, which was much higher than those from ZnO-PPH# (46.7%), ZnO-PPB# (9.6%) and ZnO-PPR# (4.1%), respectively ( $P < 0.05$ ). It could be concluded that the introduction of coupling agent inhibited the migration of zinc into 3% acetic acid. The structure of the titanate coupling agent contains two separate districts including inorganic and organic districts. The functional group (RO)<sub>m</sub> in the inorganic district can react with hydroxyl of ZnO. The organic district can occupy different organic functional groups which not only have different chemical properties but also can react with different organic materials.<sup>20</sup> The titanate coupling agent acts as a “bridge” by linking inorganic nanoparticles and organic polymers. As a result, the “bridge” accompanying with binding force prevents the surface-modified ZnO-NPs leaving from the composite polymers into the external solvent.

The result of cycle migration test showed that the amount of total zinc migrated into the simulant at 70°C for 2 hours dramatically decreased with the increase of the cycles as shown in Table 2. The amount of zinc was much higher in the first cycle compared with that in the subsequent cycles. This phenomenon was in accordance with that reported by Echegoyen and Nerin<sup>21</sup> when they measured the Kinetic Go Green samples. It was also found that the coupling agent inhibited the migration; however, this effect lessened gradually with the cycle times. Furthermore, the ratio of the total migration amount from PPH is the highest in the first cycle, whereas in the second and third cycles the ratio from PPR was higher than that from PPB, and much higher than that from PPH. It could be inferred that there are more ZnO-NPs remaining on the surface of PPH instead of in the PPH films during the manufacture processing. The initial release of



**FIGURE 1** Migration rate (%) of zinc from different types of PP films into 3% (w/v) acetic acid at A, 20°C; B, 40°C; and C, 70°C ( $n = 3$ )

**TABLE 2** Migration of zinc from different types of nanocomposite films into 3% acetic acid in cycle-use experiments (condition: 70°C, 2 h,  $n = 3$ )

| Samples  | Initial Concentration (g·kg <sup>-1</sup> ) | Total Migration after Three Cycles (mg·kg <sup>-1</sup> , Mean ± S.D.) | Ratio of Total Migration (%) |         |         |
|----------|---------------------------------------------|------------------------------------------------------------------------|------------------------------|---------|---------|
|          |                                             |                                                                        | Cycle 1                      | Cycle 2 | Cycle 3 |
| ZnO-PPH  | 10.54                                       | 6280.0 ± 241.6                                                         | 85.9                         | 10.5    | 3.6     |
| ZnO-PPH# | 11.61                                       | 5353.4 ± 79.7                                                          | 80.4                         | 15.0    | 4.5     |
| ZnO-PPB  | 12.46                                       | 1431.9 ± 58.5                                                          | 70.0                         | 21.0    | 9.1     |
| ZnO-PPB# | 13.07                                       | 933.5 ± 39.3                                                           | 69.8                         | 20.6    | 9.6     |
| ZnO-PPR  | 11.99                                       | 595.4 ± 125.9                                                          | 70.6                         | 20.0    | 9.4     |
| ZnO-PPR# | 12.22                                       | 309.9 ± 62.4                                                           | 62.4                         | 21.6    | 16.0    |

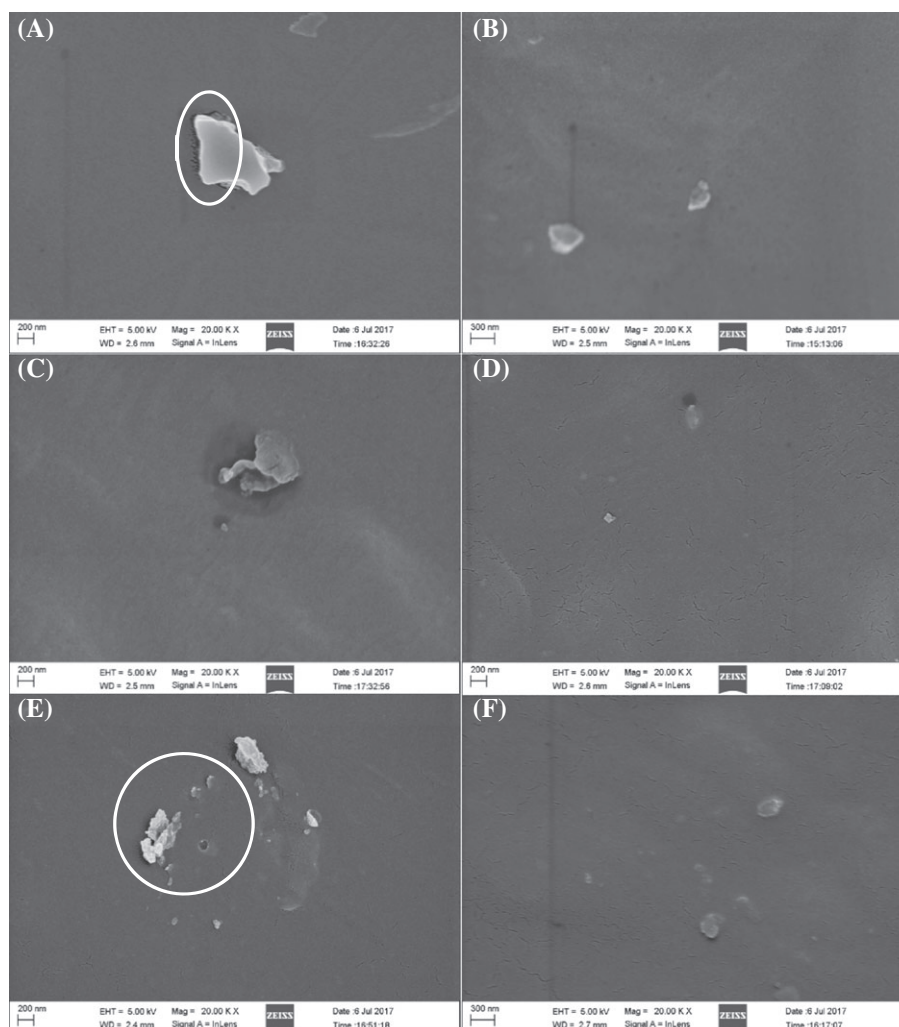
the ZnO-NPs may be because of the nanoparticles located on the surface, most of which are easily released to acetic acid as Zn<sup>2+</sup> ions.

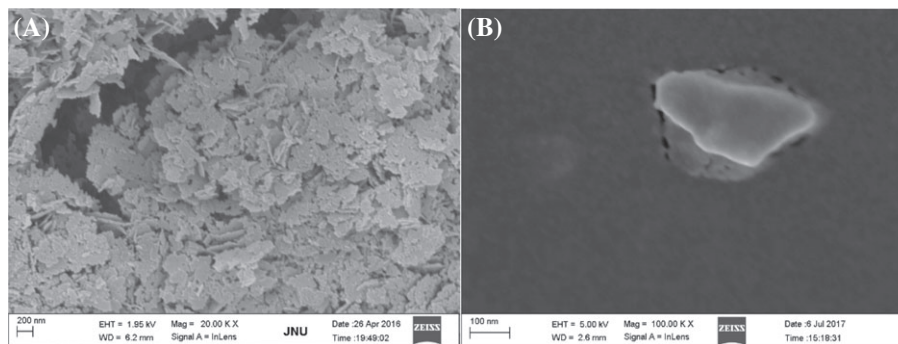
### 3.3 | Morphological studies

Figure 2 showed SEM images of three different types of nano-ZnO/PP films with or without coupling agent at 20 000× magnification. The ZnO-NPs in ZnO-PPH# film were distributed better and relatively uniform than in ZnO-PPH film. Similar results have been found in both PPB and PPR. This confirms that the coupling agent can decrease the surface energy and reduce the aggregation among particles to contribute to the uniform distribution of ZnO-NPs in the nanocomposite PP

matrix.<sup>22</sup> The original morphology of ZnO-NPs was flaky as shown in Figure 3A.

The notable phase separation could be observed from the morphology of nano-ZnO/PPH, while ZnO-NPs could be well dispersed in the PPR matrix. Figures 2A (encircled in the image) and 3B exhibited some trenches along the edge of the aggregation in ZnO-PPH film suggesting that the ZnO-NPs were more difficult to incorporate the matrix of PPH to some extent due to its high crystallinity and compact structure. In comparison, the ZnO-NPs were well embedded in both PPB and PPR films because the existence of three kinds of chains (PP chains, PE chains, and PP-PE chains) can form more voids.<sup>23</sup> These phenomena could induce that the compatibility of PP and ZnO-NPs

**FIGURE 2** FE-SEM images of A, ZnO-PPH film; B, ZnO-PPH# film; C, ZnO-PPB film; D, ZnO-PPB# film; E, ZnO-PPR film; F, ZnO-PPR# film



**FIGURE 3** FE-SEM image of A, ZnO-NPs at 20 000 $\times$  magnification; B, ZnO-PPH film at 100 000 $\times$  magnification

was PPR > PPB > PPH. This could respond to the migration behavior discussed above.

$R_a$  and  $R_q$  of all samples obtained by AFM were given in Table 3. PPB film had the highest roughness due to the "island structure."<sup>24</sup> PPR had the lowest roughness likely due to its low crystallinity and good compatibility with ZnO-NPs. The addition of coupling agent contributed the better flatness of the films.

### 3.4 | DSC analysis

Differential scanning calorimetry data for all samples before migration are summarized in Table 4. The crystallinity of PP was estimated using a heat of fusion of 209 J·g<sup>-1</sup> for 100% crystalline PP.<sup>25</sup> In other words,

the crystallinity ( $X_c$ ) of the polymer within nanocomposite film is calculated from the ratio of the current fusion enthalpy ( $\Delta H_m$ ) to the theoretical value ( $\Delta H_0$ ).

The crystallinity of the samples decreased in the order of PPH, PPB, and finally the PPR as expected. ZnO-NPs modified with coupling agent restricted the molecular chain segment of PPH and reduced its  $X_c$  from 88.6% to 73.2%. So, the migration rate of zinc was dependent on  $X_c$  for PP and the order of the maximum migration rate from PP films was PPH, PPB, and PPR. Zaman et al<sup>26</sup> considered that the addition of ZnO promoted the crystallization of PP due to heterogeneous nucleation. Zhao and Li<sup>27</sup> also found the crystallinity of PP with 3 wt% ZnO-NPs was slightly increased from 38.7% to 39.9% and no change with 1.5 wt% ZnO-NPs. However, Zhu et al<sup>28</sup> reported that the addition of nanoparticles decreased the crystallinity due to the continuity of the polymer matrix disturbed by the nanoparticles. Further studies are required to clarify the effect of nanoparticles on crystallinity and its influence on migration.

**TABLE 3** The surface roughness of sample films ( $n = 4$ )

| Samples  | $R_a$ (nm, mean $\pm$ S.D.) | $R_q$ (nm, mean $\pm$ S.D.) |
|----------|-----------------------------|-----------------------------|
| b-PPH    | 176 $\pm$ 23                | 221 $\pm$ 31                |
| ZnO-PPH  | 237 $\pm$ 13                | 300 $\pm$ 19                |
| ZnO-PPH# | 197 $\pm$ 16                | 246 $\pm$ 23                |
| b-PPB    | 316 $\pm$ 8                 | 408 $\pm$ 12                |
| ZnO-PPB  | 310 $\pm$ 50                | 407 $\pm$ 71                |
| ZnO-PPB# | 225 $\pm$ 36                | 293 $\pm$ 42                |
| b-PPR    | 157 $\pm$ 22                | 195 $\pm$ 26                |
| ZnO-PPR  | 155 $\pm$ 15                | 193 $\pm$ 19                |
| ZnO-PPR# | 145 $\pm$ 15                | 183 $\pm$ 19                |

**TABLE 4** DSC characteristics of different types of PP films

| Samples  | $\Delta H_0$ /(J·g <sup>-1</sup> ) | $\Delta H_{m1}$ /(J·g <sup>-1</sup> ) | $\Delta H_{m2}$ /(J·g <sup>-1</sup> ) | $X_c$ , % |
|----------|------------------------------------|---------------------------------------|---------------------------------------|-----------|
| b-PPH    | 209                                | 83.21                                 | 102.00                                | 88.6      |
| ZnO-PPH  | 209                                | 88.93                                 | 102.60                                | 91.6      |
| ZnO-PPH# | 209                                | 72.02                                 | 80.92                                 | 73.2      |
| b-PPB    | 209                                | 65.47                                 | 84.56                                 | 71.8      |
| ZnO-PPB  | 209                                | 64.28                                 | 75.77                                 | 67.0      |
| ZnO-PPB# | 209                                | 75.39                                 | 79.28                                 | 74.0      |
| b-PPR    | 209                                | 58.87                                 | 56.12                                 | 55.0      |
| ZnO-PPR  | 209                                | 60.23                                 | 67.85                                 | 61.3      |
| ZnO-PPR# | 209                                | 62.96                                 | 66.20                                 | 61.8      |

Note:  $\Delta H_0$  is the fusion enthalpy for a theoretically 100% crystalline PP,  $\Delta H_{m1}$  is the enthalpy of fusion in the first melting curve,  $\Delta H_{m2}$  is the enthalpy of fusion after elimination of the thermal history, and  $X_c$  is the crystallinity.

## 4 | CONCLUSIONS

The migration of zinc from nano-ZnO/PP composite films into 3% acetic acid is affected by immersion time and temperature as general. The type of PP largely influenced the migration rate of zinc, and the order of the maximum migration rate from PP films was PPH, PPB and PPR. This is a totally different conclusion compared with the organic additives which usually have the opposite order. The titanate coupling agent could inhibit the migration of zinc into 3% acetic acid. The morphology via SEM strongly supported the fact that there was a notable phase separation between ZnO-NPs and PPH. The degree of crystallinity ( $X_c$ ) for PPH decreased from 88.6% to 73.2% because the molecular chain segment of PPH was restricted by ZnO-NPs modified with titanate coupling agent. Thus, it is obvious that the utilization for titanate coupling agent and PPR as matrix could inhibit the migration of zinc. For food safety reason, PPH should be avoided as matrix while PPR was the optimized matrix when the nano-ZnO/PP composite materials are applied.

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